Do System One Models Make Better Rerankers?

Anonymous

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Abstract. System One models, or decision models, answer a question about an input with a calibrated probability. They are only a month old but much discussed, and first reports have tried them as rerankers. For six Qwen models (0.6B to 27B parameters) and ten test collections, we compare the untuned model with its decision fine-tunes and its reranker fine-tunes, one of which we trained ourselves. The reranker fine-tunes rank better wherever both exist, while decision fine-tuning adds little to the untuned model. Decision models are not better calibrated either: with a fixed probability threshold, reranker fine-tunes select better passages, and our own reranker fine-tune gives the most reliable probabilities of all systems we tested.

Keywords: Reranking · Calibration · System One Models · Fine-Tuning

1 Introduction

System One models, also called decision models, answer a question about an input (the *state*) with a probability instead of generated text, like fast, intuitive thinking [14]. TypeSafe AI released the first of them, Jev, on 15 September 2026, trained with a new method it calls RLCD (reinforcement learning for calibrated decisions) [31]. The model class is only about a month old, yet it is marketed widely, and first reports show Jev reranking BM25 candidates about as well as strong cross-encoders [11, 13, 15, 26]. Open alternatives with the same kind of output followed within weeks: Kev by the end of September [22] and Cloudflare's Clef on 1 October [4]. They fine-tune open Qwen models, but not with RLCD, which is unpublished; Kev, for instance, uses a standard cross-entropy loss [21]. Still, the largest Kev matches Jev as a reranker in our experiments. We therefore study *decision fine-tuning*, i.e. fine-tuning an open model to give Jev's kind of output, and ask, one month after Jev's release, whether it is of use for retrieval.

A fair test compares decision fine-tuning with the established alternative, reranker fine-tuning, on the same *foundation model*, i.e. the same pretrained language model of a given size, such as Qwen3.5-4B. For six Qwen foundation models from 0.6B to 27B parameters and ten test collections, we compare three variants: the base model asked zero-shot, its decision fine-tune and its reranker fine-tune. Since decision models are advertised for their calibrated probabilities, we also check whether they at least deliver that:

RQ1 On the same foundation model, does decision fine-tuning make a better reranker than reranker fine-tuning or no fine-tuning?

RQ2 Are decision models at least better calibrated than reranker fine-tunes of the same foundation model?

Both answers are negative. (1) Wherever both exist (0.6B to 9B), the reranker fine-tune ranks better than the decision fine-tune, also on collections without training data, and up to 9B decision fine-tuning hardly improves on the base model. (2) Decision models are not better calibrated than reranker fine-tunes of the same model. Our own reranker fine-tune of Qwen3.5-9B ranks as well as the best published one and is the best calibrated system we tested. Code, raw outputs and the results on all collections are available.[1]

2 Related Work

Fine-tuned rerankers. Cross-encoders fine-tuned on MS MARCO [19, 20] and their LLM successors read relevance from the score of a relevance token or a scoring head. LLM rerankers fine-tune open models such as LLaMA [17] or Qwen [34, 35]. Listwise rerankers output a ranking instead [24, 29], and the Set-Encoder lets passages attend to each other [27]. Abdallah et al. [1] compare many of them on our candidates.

System One models in retrieval. Besides Jev [31], Kev [22], Clef [4] and Laya [5] are open System One models. Their use as rerankers has so far only been tried in reports that are not peer-reviewed: S1Rank and jev-rerank score BM25 candidates with Jev [13, 26], blog posts report on Jev as a reranker [11, 15, 32], and Laya has been used as a relevance gate before recall [28]. None compares decision fine-tuning with reranker fine-tuning on the same model.

Calibration and cutoffs. Modern neural networks tend to be overconfident, and temperature scaling is a simple fix after training [12]; neural rankers are no exception [23]. Calibrated probabilities matter wherever a score is read on an absolute scale, e.g. to cut off a ranked list [2].

3 Experimental Setup

Triplets. For each of six foundation models (Qwen3-0.6B and 8B, Qwen3.5-0.8B, 4B and 9B, and Qwen3.8-27B) we compare three variants. *Base*: the released model, asked the decision models' question in its chat format (thinking off) and scored by the probability of "yes" as the first answer token. *Decision fine-tune*: Kev [21, 22] (LoRA on the pretrained checkpoints up to 9B, a full fine-tune at 27B), Clef-Flash (9B) and Clef (27B) [4]. *Reranker fine-tune*: Qwen3-Reranker [35] (on the same pretrained checkpoints as Kev) and Prism [34], scored by the probability of "yes" against "no"; none is public for 27B. For Qwen3.5-9B we add our own (*Ours-9B*), trained without any decision data: LoRA ($r = 32$),

[1] [TODO: anonymised link]

Table 1. Base model, decision fine-tune and reranker fine-tune of the same foundation model (nDCG@10, ×100, BM25 top-100). Avg: mean over all ten collections, including DL20 and DL22, which are not shown for space; ZS: mean over the collections without a training split (TREC-COVID, Touché, DBPedia). Per foundation model and column: best in bold, second best underlined. Training: ○ base model, ▲/△ decision fine-tune (joint/isolated), ■ reranker fine-tune (rows shaded grey, red, blue); *Ours-9B*: our reranker fine-tune. Paired t-test, Holm-corrected over the ten collections, $p < 0.05$: * reranker fine-tune vs. the decision fine-tune with the best Avg, † that decision fine-tune vs. the base model, ‡ joint vs. isolated. No reranker fine-tune of the 27B model exists.

			TREC DL			BEIR					Mean	
	System	Mode	DL19	DL21	DL23	COVID	NFC	SciF	Touché	DBP	Avg	ZS
Qwen3-0.6B												
○	Qwen3-0.6B		32.6	32.5	14.3	50.3	16.2	18.8	11.5	14.3	23.8	25.4
▲	Kev-0.6B	joint	18.4‡	19.1‡	11.0‡	34.0‡	12.0‡	1.1‡	4.5‡	6.1‡	13.2	14.9
△	Kev-0.6B	isolated	33.4	39.8	18.8	58.2	17.0	16.2	13.0	21.5†	27.0	30.9
■	Qwen3-Reranker-0.6B		**69.5***	**65.2***	**38.8***	**86.2***	**36.4***	**76.6***	**35.7***	**42.7***	**56.1**	**54.9**
Qwen3-8B												
○	Qwen3-8B		70.4	65.8	42.1	80.2	34.1	72.8	24.0	37.1	53.3	47.1
▲	Kev-8B	joint	63.4‡	60.1	35.7	48.2‡	19.8‡	19.2‡	9.7‡	36.7‡	38.8	31.5
△	Kev-8B	isolated	71.2	61.4†	37.6†	80.8	35.2	73.9	26.5	40.0†	52.7	49.1
■	Qwen3-Reranker-8B		**73.2**	**67.0***	**45.2***	**84.0**	**37.3***	**79.9***	**43.1***	**46.2***	**59.4**	**57.8**
Qwen3.5-0.8B												
○	Qwen3.5-0.8B		47.7	53.3	27.9	65.3	20.4	41.1	21.9	25.8	37.6	37.7
▲	Kev-0.8B	joint	35.7	38.6‡	18.2‡	45.3‡	18.6	20.2‡	9.4‡	14.6‡	25.3	23.1
△	Kev-0.8B	isolated	42.8	48.3	23.8	71.9	20.9	36.0	18.0	32.2†	36.0	40.7
■	Prism-0.8B		**72.3***	**66.3***	**43.7***	**84.2***	**36.1***	**76.3***	**35.6***	**41.7***	**57.1**	**53.8**
Qwen3.5-4B												
○	Qwen3.5-4B		69.0	66.2	40.1	79.5	34.1	72.9	24.8	37.0	52.7	47.1
▲	Kev-4B	joint	65.5	62.4	38.3	75.3‡	29.3‡	62.4‡	23.7	40.7	49.8	46.6
△	Kev-4B	isolated	62.6†	62.0	35.4	82.6	34.1	70.7	23.0	39.7†	51.3	48.4
■	Prism-4B		**74.4***	**70.3***	**46.2***	**84.6**	**37.4***	**76.9***	**34.3***	**43.7***	**58.7**	**54.2**
Qwen3.5-9B												
○	Qwen3.5-9B		69.6	67.7	38.1	81.2	33.6	75.0	27.5	35.8	53.7	48.2
▲	Kev-9B	joint	63.1	63.6	39.1	78.2	31.5	60.9‡	22.6	38.5‡	50.2	46.4
△	Kev-9B	isolated	65.7	60.8	35.4	77.7	32.8	72.6	19.8	34.1	50.1	43.8
▲	Clef-Flash	joint	68.8	64.3	40.4	82.1	35.1‡	65.4	21.3	38.8‡	52.8	47.4
△	Clef-Flash	isolated	68.8	66.4	43.8†	82.3	**37.3**†	68.7†	22.1	42.6†	54.7	49.0
■	Prism-9B		**74.0***	**71.0***	44.2	83.0	37.2	**79.8***	34.7*	43.2	59.0	53.7
■	*Ours-9B*		73.8*	67.3	**47.0**	**84.2**	37.1	77.9*	**36.5***	**47.7***	**59.4**	**56.1**
Qwen3.8-27B												
○	Qwen3.8-27B		71.6	68.5	44.1	83.0	37.6	78.5	30.2	45.0	57.4	52.7
▲	Kev-27B	joint	**73.9**	69.9	**49.3**†	**85.9**†	**38.7**†	**81.1**†	**40.8**‡	46.9†	**61.1**	**57.9**
△	Kev-27B	isolated	73.4	**70.1**	46.8	85.5	38.3	79.9	34.0	**47.5**	59.8	55.7
▲	Clef	joint	71.4	68.6	45.9	82.9	36.3	62.9‡	30.8	42.8	56.1	52.2
△	Clef	isolated	69.4	66.7	42.9	83.5	37.8	71.9	33.9	44.1	56.4	53.8

one epoch over 100k MS MARCO training queries with one relevant and 15 hard negative passages each, a ranking loss (InfoNCE) plus a yes/no classification loss (binary cross-entropy). We removed training queries that match a test or development topic and chose the checkpoint on 200 MS MARCO dev queries. Qwen3-Reranker, Prism and Ours-9B saw MS MARCO; Kev's training data contain neither MS MARCO nor BEIR [21].

Collections and candidates. TREC DL 2019–2023 [6–10] (MS MARCO v1/v2 passages [18]) and TREC-COVID, NFCorpus, SciFact, Touché and DBPedia from BEIR [30]. TREC-COVID, Touché and DBPedia have no training split in any fine-tune's data and form the *zero-shot subset* (ZS). Every system reranks the BM25 top-100 of Pyserini [16], the candidates of S1Rank [26], RankGPT [29], RankZephyr [24] and Abdallah et al. [1], whose published numbers we use where we did not rerun a system.

Decision models. All decision models get S1Rank's question [26], fixed in advance ("Is document `{doc}` relevant to the `query`? Relevant means it answers the query or contains information that directly addresses it."), with a state that holds all 100 candidates (*joint*) or one (*isolated*). Besides the triplets we run Jev 1.13.0 (closed), Laya [5] (isolated only, as its context is too short), monoT5-3B [20], bge-reranker-v2-m3 [3], the Set-Encoder [27] and RankZephyr (DL only).

Metrics. nDCG@10, with paired t-tests over queries, Holm-corrected across the ten collections ($p < 0.05$). Expected calibration error (ECE, 15 bins) [12] on the judged pairs of DL 2019–2023 (relevant from grade 2), the only collections that also judge non-relevant passages. *One cutoff*: we keep every candidate with $p \geq 0.5$ and report the F_1 of this set and its regret, i.e. how much F_1 the best cutoff per collection would add (unjudged passages count as non-relevant). *After the fact*: temperature and Platt scaling of each reranker's score, fitted on the BM25 top-100 of 1,000 MS MARCO dev queries. *Sanity checks*: our BM25 matches Pyserini's reproductions [25], monoT5-3B and RankZephyr are within 0.2 points of their published DL 2019/2020 numbers [1], and Jev joint is within 0.5 of S1Rank's [26] on four of five shared collections (1.5 below on TREC-COVID).

4 Decision vs. Reranker Fine-Tuning

Reranker fine-tuning wins on every model. For all five foundation models where both exist, the reranker fine-tune ranks better than the decision fine-tune in either mode (Table 1; average over ten collections, Qwen3-0.6B: 56.1 vs. 27.0; Qwen3-8B: 59.4 vs. 52.7; Qwen3.5-0.8B: 57.1 vs. 36.0; Qwen3.5-4B: 58.7 vs. 51.3; Qwen3.5-9B: 59.0 for Prism and 59.4 for Ours-9B vs. 54.7 for Clef-Flash). The difference is significant on 46 of the 60 pairs of model and collection, and the decision fine-tune is never significantly better (*). The zero-shot subset shows the same (e.g. 54.9 vs. 30.9 at 0.6B, 56.1 vs. 49.0 at 9B), so the gap does not come from training on MS MARCO. It does not come from special training data either, as Ours-9B, trained on MS MARCO alone, matches Prism-9B. Up to 9B, decision fine-tuning hardly changes the base model: its better mode lies between 3.5 points below and 3.2 points above the base average, with significant differences on at most five collections (†). Reranker fine-tuning adds 5 to 32 points. Only Kev-27B clearly improves on its base model (by 3.7 points, significant on eight collections). It is the best system in our study (61.1), but only slightly ahead of reranker fine-tunes a third of its size (59.4).

Table 2. Other systems, same candidates (nDCG@10 ×100, means). †: includes published numbers [1,33]; –: not all collections available. Best in bold, second best underlined; symbols and shading as in Table 1.

	System	Type	DL	BEIR	Avg	ZS
	BM25		39.3	47.1	43.2	45.2
■	RankGPT-4	listwise	–	56.9†	–	57.1†
■	RankZephyr-7B	listwise	61.4	53.3†	57.3†	51.7†
■	monoT5-3B	cross-encoder	60.4	54.6†	57.5†	52.5†
■	bge-reranker-v2-m3	cross-encoder	60.4†	52.8†	56.6†	52.2†
■	Set-Encoder-large	cross-encoder	**63.6**†	54.4	59.0†	53.8
▲	Jev	System One, joint	62.7	**58.2**	**60.5**	**57.2**
△	Jev	System One, isolated	61.4	57.6	59.5	56.7
△	Laya	System One, isolated	47.3	40.7	44.0	43.7

Joint vs. isolated. Scoring all candidates together helps only models trained on long states, and only a little: Kev-27B gains 1.3 points and Jev 1.0 (Table 2). Kev-9B and Clef stay within 0.3, Kev-4B and Clef-Flash lose 1.5 to 2, and Kev-0.6B, Kev-0.8B and Kev-8B lose 10.7 to 13.9 points, significantly on 6 to 10 collections (‡). Kev-0.8B and Kev-4B were trained on states of at most 7,552 tokens and Kev-27B on up to 32,768 [21] (Kev-8B's card gives no length), while a joint state has about 13k tokens on TREC DL and 40k or more on BEIR. Beyond the trained length the model loses track of which passage a question is about: Kev-0.6B in joint mode separates relevant from non-relevant passages only at chance level on DL 2019/2020.

Other systems. The open Kev-27B (61.1) matches the closed Jev (60.5, Table 2). Both are ahead of established rerankers (Set-Encoder-large 59.0, monoT5-3B 57.5), and so are Qwen3-Reranker-8B and Ours-9B (59.4). Laya (44.0) is barely above BM25 (43.2).

5 Are Decision Models Better Calibrated?

A model is well calibrated if its probabilities match how often passages are actually relevant. We report two measures (Table 3). The expected calibration error (ECE) measures only this match. The Brier score, the mean squared error of the probabilities, also rewards a model that separates relevant from non-relevant passages well. Calibration is the main selling point of decision models, and the base models are indeed poorly calibrated (ECE up to 0.750). Decision fine-tuning improves both measures for every model, but so does reranker fine-tuning. Compared with the reranker fine-tune of the same model, the decision fine-tune has a higher Brier score at 0.6B, 0.8B, 4B and 8B (e.g. Kev-4B 0.207 vs. Prism-4B 0.172) and a higher or similar ECE (Kev-4B 0.216 vs. 0.218). Only at 9B is Clef-Flash better than Prism (ECE 0.140 vs. 0.242), and there Ours-9B has the lowest ECE of the study (0.111, below every decision model including the 27B ones and Jev) and a Brier score as low as the best decision model (0.157).

Table 3. Calibration on the judged pairs of TREC DL 2019–2023 (means; lower is better). ECE: expected calibration error (15 bins); Brier: mean squared error of the probability against the 0/1 label. Per foundation model: base model, decision fine-tunes in their better-calibrated mode (j/i: joint/isolated), reranker fine-tunes; best in bold, second best underlined. Symbols and shading as in Table 1.

	System	ECE	Brier
Qwen3-0.6B			
○	Qwen3-0.6B	0.750	0.750
△	Kev-0.6B (i)	<u>0.579</u>	<u>0.543</u>
■	Qwen3-Reranker-0.6B	**0.511**	**0.481**
Qwen3-8B			
○	Qwen3-8B	0.383	0.380
▲	Kev-8B (j)	**0.243**	<u>0.248</u>
■	Qwen3-Reranker-8B	<u>0.257</u>	**0.238**
Qwen3.5-0.8B			
○	Qwen3.5-0.8B	0.716	0.694
△	Kev-0.8B (i)	<u>0.357</u>	<u>0.309</u>
■	Prism-0.8B	**0.243**	**0.193**
Qwen3.5-4B			
○	Qwen3.5-4B	0.296	0.269
△	Kev-4B (i)	**0.216**	<u>0.207</u>
■	Prism-4B	<u>0.218</u>	**0.172**

	System	ECE	Brier
Qwen3.5-9B			
○	Qwen3.5-9B	0.283	0.265
▲	Kev-9B (j)	0.237	0.209
▲	Clef-Flash (j)	<u>0.140</u>	<u>0.165</u>
■	Prism-9B	0.242	0.183
■	*Ours-9B*	**0.111**	**0.157**
Qwen3.8-27B			
○	Qwen3.8-27B	0.166	0.180
▲	Kev-27B (j)	<u>0.156</u>	<u>0.159</u>
▲	Clef (j)	**0.147**	**0.157**
Other systems			
▲	Jev (j)	**0.224**	**0.187**
■	monoT5-3B	0.312	0.295
■	bge-reranker-v2-m3	<u>0.231</u>	<u>0.228</u>
■	Set-Encoder-large	0.495	0.426

Good calibration thus comes from a loss that treats the score as a probability, not from decision fine-tuning.

Uses of calibration. The same picture holds for practical uses of the probabilities (results in our repository). If every passage with $p \geq 0.5$ is kept, e.g. to decide which passages reach a generator, the best reranker fine-tune selects the better set for every model where both exist, and a fixed threshold works across collections for Prism and bge as well as for the decision models. Summing a query's probabilities estimates its number of relevant passages about equally well for decision and reranker fine-tunes, and best for Ours-9B. Calibrating a reranker after training with temperature or Platt scaling on MS MARCO dev does not reach the decision models, because MS MARCO judges only about one passage per query. A well calibrated model is better, but it need not be a decision model, as Ours-9B shows.

6 Conclusion

Do System One models make better rerankers? No. On the same foundation model, reranker fine-tuning beats decision fine-tuning, which adds little to the untuned model. Their calibration is no advantage either: reranker fine-tunes are as well calibrated, select better passages at a fixed threshold, and can be trained to be better calibrated than any decision model. Despite the hype and first uses as rerankers, decision models are currently of no use for reranking, neither for ranking nor for their calibration.

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